

Chapter 1 Energy and Matter

Every living cell is separated from its surroundings by a thin boundary that controls what enters and what leaves. This boundary is not a rigid wall but a flexible, self-sealing film that constantly rearranges itself as the cell grows, divides and responds to signals.

1.1 Systems and Surroundings

Researchers first described the boundary as a simple film of oil, but careful measurements in the twentieth century showed that it has a defined thickness and a layered structure. Those measurements led to the models that are still taught in introductory courses today.

Open systems

Proteins embedded in the boundary act as channels, pumps and receptors. Some span the full width of the film, while others sit on only one face. Their arrangement explains why the two faces of the boundary behave differently.

Isolated systems

Small, uncharged molecules such as oxygen cross the boundary easily, whereas ions and large molecules need help. The cell spends a large share of its energy budget on moving these substances against their natural direction of flow.

1.2 Systems and Surroundings Revisited

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Closed systems

Proteins embedded in the boundary act as channels, pumps and receptors. Some span the full width of the film, while others sit on only one face. Their arrangement explains why the two faces of the boundary behave differently.

Boundaries

Small, uncharged molecules such as oxygen cross the boundary easily, whereas ions and large molecules need help. The cell spends a large share of its energy budget on moving these substances against their natural direction of flow.